

Volume 5 – REST API & WebSocket Specification

AI Quant Trading Platform

Version: 1.0

API Version: v1

Document Type: API Specification

1. Introduction

This document defines all external and internal APIs for the AI Quant Trading Platform.

Goals:

- Consistent REST design
- Versioned APIs
- Secure authentication
- Predictable responses
- WebSocket support for real-time data
- Idempotent write operations where required

Base URL

```
https://api.domain.com/api/v1/
```

WebSocket

```
wss://api.domain.com/ws/
```

2. Authentication

Authentication Method

- JWT Access Token
- Refresh Token

Headers

```
Authorization: Bearer <access_token>
```

Protected APIs require authentication unless explicitly marked public.

3. Standard Response Format

Success

```
{
  "success": true,
  "message": "Operation completed.",
  "data": {},
  "meta": {}
}
```

Error

```
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Input validation failed."
  }
}
```

4. Authentication APIs

POST /auth/register

POST /auth/login

POST /auth/logout

POST /auth/refresh

POST /auth/forgot-password

POST /auth/reset-password

POST /auth/change-password

POST /auth/verify-email

POST /auth/enable-2fa

POST /auth/disable-2fa

GET /auth/profile

PATCH /auth/profile

DELETE /auth/account

5. User APIs

GET /users/me

PATCH /users/me

GET /users/preferences

PATCH /users/preferences

GET /users/security

PATCH /users/security

GET /users/activity

6. Exchange APIs

GET /exchanges

GET /exchanges/{id}

POST /exchange-accounts

GET /exchange-accounts

PATCH /exchange-accounts/{id}

DELETE /exchange-accounts/{id}

POST /exchange-accounts/{id}/test

POST /exchange-accounts/{id}/sync

GET /exchange-accounts/{id}/balances

GET /exchange-accounts/{id}/positions

GET /exchange-accounts/{id}/orders

7. Market APIs

GET /markets

GET /markets/{symbol}

GET /markets/{symbol}/ticker

GET /markets/{symbol}/orderbook

GET /markets/{symbol}/ohlcv

GET /markets/{symbol}/trades

GET /markets/{symbol}/funding-rate

GET /markets/{symbol}/open-interest

GET /markets/search

8. Scanner APIs

POST /scanner/run

GET /scanner/status

GET /scanner/results

GET /scanner/candidates

GET /scanner/rejected

GET /scanner/history

PATCH /scanner/settings

9. Technical Analysis APIs

GET /indicators

GET /indicators/{symbol}

GET /indicators/{symbol}/ema

GET /indicators/{symbol}/rsi

GET /indicators/{symbol}/macd

GET /indicators/{symbol}/atr

GET /indicators/{symbol}/trend

GET /indicators/{symbol}/support-resistance

10. AI APIs

POST /ai/analyze

GET /ai/score/{symbol}

GET /ai/recommendation/{symbol}

GET /ai/explanation/{symbol}

GET /ai/history

GET /ai/models

POST /ai/models/retrain

GET /ai/performance

11. Strategy APIs

GET /strategies

POST /strategies

GET /strategies/{id}

PATCH /strategies/{id}

DELETE /strategies/{id}

POST /strategies/{id}/activate

POST /strategies/{id}/deactivate

POST /strategies/{id}/backtest

12. Risk APIs

GET /risk/profile

PATCH /risk/profile

GET /risk/exposure

GET /risk/limits

PATCH /risk/limits

GET /risk/history

POST /risk/emergency-stop

POST /risk/resume

13. Trading APIs

POST /orders

GET /orders

GET /orders/{id}

PATCH /orders/{id}

DELETE /orders/{id}

GET /positions

GET /positions/{id}

POST /positions/{id}/close

POST /positions/{id}/partial-close

GET /trades

GET /trade-history

14. Portfolio APIs

GET /portfolio

GET /portfolio/assets

GET /portfolio/history

GET /portfolio/performance

GET /portfolio/pnl

GET /portfolio/allocation

15. Reports APIs

GET /reports

POST /reports/generate

GET /reports/{id}

DELETE /reports/{id}

GET /reports/download/{id}

16. Notification APIs

GET /notifications

PATCH /notifications/read

DELETE /notifications/{id}

GET /notification-settings

PATCH /notification-settings

17. Paper Trading APIs

POST /paper/start

POST /paper/stop

GET /paper/orders

GET /paper/positions

GET /paper/performance

DELETE /paper/reset

18. Backtesting APIs

POST /backtest

GET /backtest/jobs

GET /backtest/{id}

DELETE /backtest/{id}

POST /backtest/optimize

19. Admin APIs

GET /admin/users

PATCH /admin/users/{id}

DELETE /admin/users/{id}

GET /admin/system

GET /admin/queues

GET /admin/workers

GET /admin/logs

PATCH /admin/settings

20. WebSocket Channels

Public Channels

market.ticker

market.trades

market.orderbook

market.ohlcv

market.funding

Private Channels

portfolio.balance

portfolio.positions

orders.status

orders.executions

notifications

scanner.results

ai.recommendations

21. WebSocket Event Format

```
{  
  "event": "market.ticker",  
  "timestamp": "2026-06-29T12:00:00Z",  
  "payload": {}  
}
```

22. Pagination

Parameters

- page
- page_size
- sort
- order

Example

```
GET /orders?page=1&page_size=50
```

23. Filtering

Supported Query Parameters

- symbol
- strategy
- exchange
- status
- date_from
- date_to

Example

```
GET /orders?symbol=BTCUSDT&status=OPEN
```

24. Rate Limits

Authentication APIs

- 10 requests/minute

Trading APIs

- Configurable per user tier

Market APIs

- Cached where applicable

Admin APIs

- Restricted to authorized roles
-

25. Error Codes

- AUTHENTICATION_FAILED
- AUTHORIZATION_DENIED
- VALIDATION_ERROR
- RESOURCE_NOT_FOUND
- EXCHANGE_UNAVAILABLE

- RATE_LIMIT_EXCEEDED
 - ORDER_REJECTED
 - RISK_LIMIT_EXCEEDED
 - INTERNAL_SERVER_ERROR
-

26. API Versioning

Current Version

v1

Future versions

v2

v3

Breaking changes must use a new API version.

27. Idempotency

Endpoints that create orders or trigger actions should support an `Idempotency-Key` header to prevent duplicate execution during retries.

28. API Documentation

Provide interactive documentation using OpenAPI/Swagger.

Deliverables:

- OpenAPI specification
 - Swagger UI
 - Postman collection
 - Example requests and responses
 - Authentication guide
 - Error reference
-

29. Integration Rules

- All APIs return JSON.
- Time values use UTC in ISO 8601 format.
- Monetary values use decimal types.

- Symbol names follow exchange conventions.
- Sensitive fields are never returned in responses.
- Deprecated endpoints remain supported for one major version with clear migration guidance.